

Instructor: Date:

Learner: Regional Branch:

Program: Center:

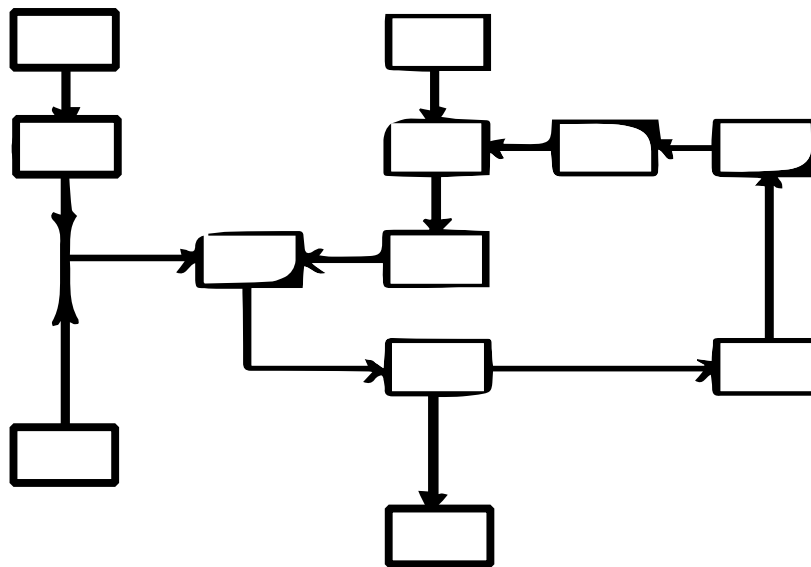
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Let's Start!

OOP Features for Data Manipulation



Learning Outcome:

Identify and explain the characteristics of software.

Discuss information about designing principles in software programming

Ask your partner:

- Do you think OOP is easy to use? Why?
- What is a paradigm in programming?

Can Do Statements:

- I can list the basic concepts in object oriented programming.
- I can mention characteristics for abstraction and encapsulation in object oriented programming.
- I can talk about characteristics for inheritance and polymorphism in object oriented programming.



Let's Explore!

Sara wants to learn more about concepts in object oriented programming.



Instructions: Read the text below.

Teacher: Guys, today we are talking about concepts in object-oriented programming or OOP concepts. How do we implement these concepts?

Sarah: We need some concepts. Those concepts are implemented by the use of classes, methods, and objects.

Teacher: You're right, but, before covering that topic, can you tell us what the four core concepts of object-oriented programming are?

Sarah: Yes, I know they are the abstraction, encapsulation, inheritance and polymorphism. However, I don't understand the difference between them.

Teacher: Don't worry, we need to take our time to understand those concepts. Let's start with abstraction. Abstraction is useful for programmers since it conceals the less important details of an object.

Sarah: It means it doesn't show the whole information but just the more relevant details. Am I right?

Teacher: You are! Of course, it eases the process and makes it more efficient. Encapsulation, on the other hand, keeps data away from danger. It doesn't allow any misuse or interference.

Sarah: How does it keep it safe?

Teacher: It hides the internal mechanism and data structures, which means it doesn't endorse any arbitrary access. It behaves as a wrapper that covers the code and data. Are these two concepts clear?

Sarah: Yes, you can continue.

Teacher: Great. Now let's talk about inheritance and polymorphism. Inheritance is a process that creates new classes from existing ones.

Sarah: Do these new classes inherit something from the predecessor ones?

Teacher: Yes! They inherit certain characteristics. We are going to cover the inheritance topic later. Let's continue with polymorphism.

Sarah: What does polymorphism mean?

Teacher: Basically, it means that something has one name and at the same time multiple forms. Talking about programming, all the properties of an object are susceptible to change their form according to a certain context.





Let's Practice!

Features on OOP

Instructions: Use the words from the word bank to complete each sentence.

- | | | |
|---------------------------------------|---------------------------------------|--|
| 1. <input type="text" value="these"/> | 2. <input type="text" value="these"/> | 3. <input type="text" value="With these"/> |
| <input type="text" value="this"/> | <input type="text" value="those"/> | <input type="text" value="With that"/> |
| <input type="text" value="that"/> | <input type="text" value="this"/> | <input type="text" value="With this"/> |

1. We implement _____ concepts according to their use of classes, methods, and object.
2. We need to use abstraction in _____ system.
3. When you study object-oriented programming, you have to learn how to work _____ concepts encapsulation, inheritance, and polymorphism.



Let's Practice!

Basic Concepts on Abstraction and Encapsulation

Instructions: Circle the correct answer to complete the paragraph.

According to the teacher, most of the programmers use the abstraction concept since it (don't show / doesn't show / shows) the whole information. On the other hand, the Encapsulation concept (allows / don't allow / doesn't allow) any misuse or interference. Thus, the encapsulation technique doesn't (hide / show / delete) all the information, just the one that doesn't need to be there.



Extension Activity

What can you say about concepts in OOP?

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about OOP concepts.
- Write four steps about how to implement OOP concepts.
- Use the demonstratives as adjectives and pronouns in your sentences.
- Explain the steps to your instructor and classmates.

Self-Assessment

- | | Yes | No |
|---|-----------------------|-----------------------|
| 1. I look for additional information about OOP concepts. | ☐ | ☐ |
| 2. I write four steps about how to implement OOP concepts. | ☐ | ☐ |
| 3. I use demonstratives as adjectives and pronouns in my sentences. | ☐ | ☐ |
| 4. I explain the steps to my instructor and classmates. | ☐ | ☐ |



Target Words

- **inherit** (verb) to take or receive things or characteristics from a predecessor
e.g. Do these new classes inherit something from the predecessor ones?
- **endorse** (verb) to approve or sustain
e.g. Encapsulation hides the internal mechanism and data structures, which means it doesn't endorse any arbitrary access.
- **wrapper** (noun) something that covers or protect certain item
e.g. Encapsulation behaves as a wrapper that covers the code and data.
- **conceal** (verb) to hide
e.g. Abstraction is really useful for programmers since it conceals the less important details of an object.
- **ease** (verb) to make something easy
e.g. Abstraction eases the process and makes it more efficient.
- **misuse** (noun) the incorrect use of something
e.g. Encapsulation doesn't allow any misuse or interference.



Learning Points

Demonstratives as Pronouns, Adjectives, in Plural Forms

Demonstratives are words we use to refer to the position of an object in relation to us, the speakers.

We can divide demonstratives into **demonstrative adjectives** and **demonstrative pronouns**.

The demonstrative pronouns are **this** (near) and **that** (far), for singular nouns; **these** (near) and **those** (far), for plural nouns. They represent and identify an object, and they can stand alone.

For example:

*I can't believe **this**!*

***This** sounds like a good idea!*

Demonstrative adjectives, on the other hand, modify nouns, which means they can't stand alone, they need a noun.

For example:

***These codes** are necessary for programming (**these** modifies **codes**).*

Observe the following chart to remember demonstratives use:

	Near the speaker	Far from the speaker
Singular	This	That
Plural	These	Those

Examples

- *Let's start with abstraction. **This** is useful for programmers since it conceals the less important details of an object.*
- *You're right, but **that's** a topic we are covering later on.*





Learning Points

Simple Present Tense in Negative Statements

You use the **Simple Present Tense** to talk about habits and general facts.

We use the present simple tense to express routines and general facts that happen on a regular basis.

In order to express a negative idea in the present simple tense, use the following structure:

Subject+ auxiliary verb (do/ does) + not+ main verb in base form+ complement.

For example:

*I **don't understand** that programming concept.*

*Jane **doesn't know** what OOP stands for.*

*The software designer of the company **doesn't like** to use open source programs to code.*

As you can see, you keep the verb in its base form, since we include the auxiliary in the sentence. Also, note that the auxiliary verb **do** changes to **does** when you use it with a third person in singular form (he, she, and it).

Examples

- *I know they are abstraction, encapsulation, inheritance, and polymorphism. Though I don't understand the difference between them.*
- *Excuse me. Does It mean it doesn't show the whole information but just the more relevant details?*